

PESTICIDES

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19.1 DEFINITION

Pesticides are substances used by human beings to kill or deter organisms (pests) that threaten our health and wellbeing, the health and wellbeing of pets and livestock, or cause damage to crops. Antibiotics in the medical sense are excluded, but included are insecticides, herbicides, fungicides, acaricides, nematocides, molluscicides and rodenticides, among others. Of these, insecticides (to control insects) and herbicides (for controlling unwanted vegetation) are used in the largest quantities¹ and have the greatest impact on the environment.

19.2 HISTORICAL BACKGROUND

Since pesticides are by definition toxic to living organisms, there is great public concern over the effect of these substances, not only on human beings, but also on non-target organisms in the environment. People fear that they themselves may be poisoned by the indiscernible residues on the food that they eat or in the air that they breathe. They are also concerned that birds, fish, and other beneficial organisms may be adversely affected by pesticides. In the 1960s, this fear was articulately expressed by Rachel Carson in her best-selling book, *Silent Spring*,² widely credited with giving impetus to the quest for the safer use of pesticides in particular and to the environmental movement in general.

Insecticides

The insecticidal activity of some naturally occurring compounds such as nicotine and pyrethrum, derived respectively from the tobacco plant and certain chrysanthemum species, has been known for centuries. These have since been supplemented by inorganic compounds for insect control and sulphur- and copper-based compounds for fungus control.

In the years between 1939 and 1953 tremendous advances were made in insect control as a result of the discovery of the insecticidal activity of a number of synthetic organic compounds such as organochlorines like dichlorodiphenyltrichloroethane (DDT), benzene hexachloride (BHC) (already synthesised as chemicals in the previous century), dieldrin and toxaphene; the organophosphates (parathion and others) and the carbamates (such as carbaryl). Only much later, in 1978, did a powerful new group, the synthetic pyrethroids, with their

¹ London & Meyers 'General Patterns of Agricultural Chemical Usage in the Southern Regions of South Africa' (1995) 91 *South African Journal of Science* 508.

² *Silent Spring* (Houghton Mifflin 1962).

characteristics of low acute mammalian toxicity and broad-spectrum effectiveness appear on the market. This was followed by a chemically related group of compounds, the chloronicotinyls, of which imidachloprid is most widely used. The latter is highly systemic, with low mammalian toxicity. Recently a number of new products with various unrelated chemical structures have also reached the market.

These organic compounds kill insects not only when ingested like the inorganic compounds, but also after absorption through the skin, ie on contact, when organisms are hit by a droplet of spray or when they walk across a sprayed surface. DDT proved to be extremely successful in the control of insect vectors of disease, such as lice, during World War II and it seemed that their total elimination was at hand. Also, tremendous increases in agricultural production were foreseen through the elimination of insect damage to crops and livestock. However, the enthusiasm over the benefits brought about by these chemicals was soon dampened by the appearance of DDT resistance in houseflies in 1946 and recognition of the ecological problems that might be caused by the widespread use of DDT and other persistent insecticides. In South Africa DDT was withdrawn as a stock remedy in 1971 and from all agricultural use in 1976. It was banned in 1983 and in terms of the Stockholm Convention of 2004 on Persistent Organic Pollutants (to which South Africa is a party) it is listed as a Persistent Organic Pollutant which may only be used by governments for malaria vector control.

Herbicides

The immediate post-war years also saw the discovery of the phenoxy herbicides, primarily 2,4-D. The selective action of this herbicide enabled farmers to control broad-leaved weeds among their cereal crops without damaging the crops. Subsequently, a great variety of herbicides with a wide range of chemical structures and with many different characteristics, most of them with a relatively low toxicity to human beings, have been developed. However, the detection of their residues in soil and water, evidence of potential health hazards caused by some of them (such as the impurity dioxin in 2,4,5-T, now withdrawn, and paraquat³), and their possible effect on rare plants have raised public concern since the late 1980s and early 1990s.

Fungicides

Fungicides are a third major group of agricultural pesticides, though the quantities used are smaller than the insecticides and herbicides. Some of them, such as sulphur and copper sulphate, have been in use for more than two centuries and are still on the market, but many new compounds have been developed. Their acute toxicity to human beings and wildlife is generally low and as a result there is less concern about their use than that of insecticides. However, discoveries about the reproductive toxicity and endocrine disruption in animals of commonly used fungicides have caused concern.⁴

³ Madely *Paraquat-Syngenta's Controversial Herbicide* Report written for the Berne Declaration, April 2002.

⁴ *Trouble on the Farm. Growing up with Pesticides in Agricultural Communities* Natural Resources Defence Council Report (1991).

19.3 ENVIRONMENTAL IMPACT

Detailed studies of the chemistry and toxicity of pesticides have since the 1990s been undertaken, and the reader is referred to two studies in particular for an in-depth analysis of the many aspects and implications of pesticide use. Tomlin's *The Pesticide Manual*⁵ is a comprehensive reference source on the active ingredients used for the control of crop pests, diseases and weeds, etc and Yu's *The Toxicology and Biochemistry of Insecticides*⁶ is a comprehensive and up-to-date review of insect toxicology, including the classification, formulation and environmental impact of insecticides.

Rather than repeating the information and comments in these publications, an attempt will be made here to approach the subject from a more ecological perspective. The aim is to explain the impact of pesticides on the different higher levels of organisation which ecologists recognise as giving structure to the great diversity found in nature. These levels include the individual, the population, the community, and the ecosystem.

19.3.1 Impact at the individual level

Two types of effect on the individual organism can be distinguished, ie an acute and a chronic effect.

Acute effect

The acute effect is the effect within seconds, minutes or hours after intake, ie the immediate effect on the organism, usually manifested after exposure to a single large dose. The acute oral and dermal toxicity of all pesticides for mammals is established during product development stages (usually by testing on laboratory rats or rabbits), long before their registration and release onto the market. Such toxicity values may be found in the lists of registered pesticides published from time to time by the Department of Agriculture⁷ or in *The Pesticide Manual*.⁸

The toxicity is expressed as an LD₅₀ (lethal dosage) value, which refers to the dosage in milligrams of pesticide per kilogram of body weight of the test animal that is lethal to 50 per cent of the test population. This value ranges from less than one for extremely toxic pesticides to more than 5 000. The value gives only a general indication of the toxicity of a pesticide for it may vary according to species, sex, and age group as well as across the animal kingdom. Hence mammals, reptiles, birds or amphibians may suffer different effects from the same chemical at the same dosage. It bears no relation to the toxicity of the chemical against insects. In general, the ideal for pesticide companies is to find chemicals that are toxic to organisms which are defined as pests, such as the case of a herbicide which specifically targets noxious or toxic plants like *Datura ferox*, but that have little or no effect on other organisms, particularly beneficial

⁵ Tomlin *The Pesticide Manual* 14 ed (British Crop Protection Enterprises 2006).

⁶ Yu *The Toxicology and Biochemistry of Insecticides* (CRC Press 2008).

⁷ Nel et al *A Guide for the Control of Plant Pests* 39 ed (2002); Anon *A Guide to the Use of Herbicides* 18 ed (Dept of Agriculture, Directorate: Food Safety and Quality Control 2004).

⁸ Tomlin loc cit.

organisms and human beings. Similarly it is desirable to target destructive insects but to have no effect on beneficial insects such as plant pollinators.

The labels on pesticide containers give an indication of the toxicity of the contents by words such as 'very toxic', 'toxic', 'harmful' and 'caution', with matching pictograms. The Global Harmonised System of Classification and Labelling of Chemicals (GHS) is a UN-supported initiative, aimed at implementing an internationally uniform system of hazard classification and labelling by 2008.⁹ South Africa took the first step towards implementing the GHS by issuing a National Standard (SANS 10234:2007) in February 2007 which adopts the GHS classification, labeling, safety data sheet (SDS) and transport criteria. This standard is in addition being used as the basis for the development of a harmonized SADC standard by the SA Community Standardization Body (SADCSTAN). A review of legislation and policy instruments was undertaken in 2007 to ensure alignment with the GHS requirements. The finalized draft legislation was due to be presented for ministerial approval in August 2008 and work is currently underway to develop a list of chemicals classified in terms of the GHS to facilitate implementation.¹⁰

Pesticides kill in many different ways. The most commonly used insecticides are nerve poisons, ie they interfere with the transmission of nerve impulses from one nerve ending to another or to innervated tissues. When the transmission of nerve impulses in the brain of higher animals is impaired, respiration is depressed and death from suffocation results. Similarly, pesticides may affect insects by interference with the sodium channel.¹¹

Acute pesticide poisoning is recognised as a global health issue, with the World Health Organisation estimating that three million severe human poisonings occur annually, of which about ten per cent result in death (the majority being suicides).¹² In South Africa the poisoning of human beings by pesticides must be notified in terms of the Health Act 63 of 1977. In the Western Cape alone, 225 cases of pesticide poisoning were notified from 1987 to 1991 and there is evidence of under-reporting.¹³ In 2002, 482 acute pesticide poisoning cases and 24 pesticide-related deaths were reported to the National Department of Health but the true rates of poisoning are estimated to be up to 20 times higher, and poisonings among women and from occupational exposure are believed to be particularly under-reported.¹⁴ Latest available statistics from the National Department of Health indicate that during the years 2001 to 2005, 2 462 cases

⁹ <http://www.epa.gov/pesticides/international/globalharmon.htm>

¹⁰ www.unece.org/trans/doc/2008/ac10c4/UN-SCEGHS-15-inf23e.doc

¹¹ Laped, Grolleau & Satelle 'Indoxacarb, An Oxadiazine Insecticide, Blocks Insect Sodium Channels' (2001) 132 *British Journal of Pharmacology* 587.

¹² Konradsen 'Acute Pesticide Poisoning — A Global Public Health Problem (2007) 54 *Danish Medical Bulletin* 58.

¹³ London et al 'Notification of Pesticide Poisoning in the Western Cape, 1987–1991' 1994 (84) *South African Medical Journal* 269.

¹⁴ Rother, Hall & London 'Pesticide Use Among Emerging Farmers in South Africa: Contributing Factors and Stakeholder Perspectives' (2008) 25 *Development Southern Africa* 4.

and 72 deaths country-wide were reported to the Department of Health.¹⁵ These figures include cases of suicide.

The acute effect of pesticides causes the death of many non-target organisms, be they mammals, birds, fish or insects. Vultures are often killed when poison is placed by farmers in stock carcasses in an attempt to poison carnivores. Guinea fowl, blue cranes and many other bird species may be killed when wheat or maize lands are sprayed or seeds treated with insecticides. The Endangered Wildlife Trust's Wildlife Conflict Prevention Group has a database of wildlife poisonings dating from 1995 to the present, and a study conducted in 1994 indicated that up to 470 000 game birds and waterfowl are poisoned annually by pesticides in South Africa.¹⁶

The acute effects of pesticide use in South Africa are also manifested in the increasing illegal use of pesticides to commit crime; for example, burglars have been known to use pesticides to poison guard dogs, poachers to kill wildlife and members of the public to kill baboons. In response to the high number of poisoning cases and the realisation that there is a lot of poison in the country that is not properly controlled, the Chemical Crime Management Forum was established in 2006, consisting of members of South African Police Services, the Endangered Wildlife Trust's Wildlife Conflict Prevention Group, the National Council of SPCAs, the Department of Agriculture, the Department of Environmental Affairs and Tourism, and pesticide manufacturers.¹⁷ The Forum will address issues around the illegal possession and misuse of poison through sharing information on hazardous substances, discussing investigations and collaborating in the investigation of wildlife and domestic poisoning cases.¹⁸

Chronic effect

The chronic or cumulative effect of continuous exposure to low levels of pesticides is less well known than the acute (immediate) effects, and is often discovered only after the product has been in use for many years. Many studies¹⁹ have indicated that some of the organochlorine compounds, particularly DDT and dieldrin, affect the reproduction of birds by inhibiting calcium absorption by hen birds, resulting in eggshell thinning and threatened egg viability. The viability of the progeny was also compromised, because hatchlings were deformed or suffered skeletal damage in early life. These compounds are also lipophilic (stored in body fats in direct proportion to the level of intake) and concentrated as they move up the food chain from plants to herbivores and carnivores, rendering those at the top of the feeding pyramid particularly vulnerable.

Many cases have been reported in world literature of pesticide residues, particularly of the organochlorine insecticides, being found in human beings and

¹⁵ <http://www.doh.gov.za/facts/stats.notes/2005/pesticides.pdf>.

¹⁶ Berutti, Snow, & van Zijl 'Deliberate Poisoning: The Biggest Threat to Gamebirds' (2005) 11 *Wingshooter* 12–16.

¹⁷ http://www.servamus.co.za/index.php?option=com_content&task=view&id=26&Itemid=9.

¹⁸ <http://www.ewt.org.za>

¹⁹ Bouwman et al 'Organochlorine Contaminants in Cormorant, Darter, Egret and Ibis Eggs from South Africa' (2007) 12 *Chemosphere*.

other mammals, birds and fish, including in South Africa,²⁰ and the decline in numbers of certain birds of prey can be attributed to chronic poisoning by organochlorine compounds. In a study of indigenous fish species in South African rivers, the maximum whole fish load of organochlorines reached high-risk levels in worst-case scenarios (that is, fish eaten daily all year round).²¹

Vultures from different localities in South Africa had low, but quantifiable residues of 14 persistent chlorinated hydrocarbon pollutants in their tissues, including BHC, chlordane, dieldrin, and endosulfan.²² A recent analysis of eggs from African darter, cattle egret, reed cormorant and African sacred ibis, among others, found HCB, DDTs, HCHs, chlordanes and PCBs at detectable levels, with the African darter birds having the highest levels of all compounds other than chlordane.²³

As regards marine organisms, low levels of dieldrin were found in mussels and mullet along the coast near Durban,²⁴ while traces of DDT and its metabolites were detected in the blubber of all the sperm whales and most of the minke and fin whales sampled off Durban during 1974.²⁵ A 2004 study of marine waters in the Eastern Cape determined the presence of persistent organochlorine pesticides (POPs) such as DDT and its metabolites (DDD and DDEs), chlordane, hexachlorobenzene (HCB), heptachlor and endosulfan.²⁶

Studies of rural water sources have increasingly reported instances of pesticide pollution resulting from farming activities. Consistent pesticide pollution of surface-and groundwater (including drinking water) was found in three rural Western Cape agricultural districts in 1996-7, and in 2001 azinphosmethyl and endosulfan were detected in the Lourens River, the result of run-off from fruit orchards.²⁷

A monitoring programme conducted in Durban during 2004-5 in both residential and industrial areas, designed to estimate long-term levels of a wide range of pesticides in the ambient air, found comparable levels of many pesticides at all sites, including banned pesticides.²⁸ Levels of p,p'-DDT (42 ± 27 pg m⁻³)

²⁰ Bouwman, Sereda & Meinhardt 'Simultaneous Presence of DDT and Pyrethroid Residues in Human Breast Milk from a Malaria Endemic Area in South Africa' 2006 *Environmental Pollution*.

²¹ Dalvie et al 'Knowledge and Attitudes in the Rural Western Cape Towards Pesticides in Water Sources' (2004) 30 *Water SA*.

²² Van Wyk et al 'Persistent Organochlorine Pesticides Detected in Blood and Tissue Samples of Vultures from Different Localities in South Africa' 2001 (129) *Comparative Biochemistry and Physiology, Part C* 243.

²³ Bouwman et al loc cit.

²⁴ Sibbald et al 'A Limited Collaborative Investigation of the Occurrence of Dieldrin in Selected Biota in the Durban Area' (1986) 82 *South African Journal of Science* 319.

²⁵ Henry & Best 'Organochlorine Residues in Whales Landed at Durban, South Africa' (1983) 14 *Marine Pollution Bulletin* 223.

²⁶ Fatoki & Awofolu 'Levels of Organochlorine Pesticide Residues in Marine-, Surface-, Ground- and Drinking Waters from the Eastern Cape Province of South Africa' (2004) 39 (1) *Journal of Environmental Science and Health*.

²⁷ Dalvie et al 'Knowledge and Attitudes in the Rural Western Cape towards Pesticides in Water Sources' (2004) 30 (1) *Water SA*.

²⁸ Butterman et al 'Organochlorine Pesticides in Ambient Air in Durban, South Africa' (2008) 397 (1-3) *Science of the Total Environment*.

and its derivatives were relatively high and showed an unusual mixture with high levels of p,p'-DDD ($12 \pm 11 \text{ pg m}^{-3}$). Other pesticides detected and quantified included aldrin, chlordanes, hexachlorobenzene and dieldrin. Potential source areas suggest that chlordane and lindane arise from both local as well as regional sources.

Fears have grown around the harmful effects to humans of long-term exposure to pesticides, particularly in the agricultural sector,²⁹ and it has long been thought that cancer, reproductive toxicity and liver effects are potential hazards.³⁰ The organochlorine pesticides are extremely persistent and accumulate in fatty tissue. Through the process of bioaccumulation (lower amounts in the environment get magnified sequentially up the food chain), large amounts of organochlorines can accumulate in top species like humans. There is substantial evidence to suggest that DDT, and its metabolite DDE, act as endocrine disruptors, interfering with hormonal function of estrogen, testosterone, and other steroid hormones.³¹

In view of their stability, their ability to spread to remote areas, their effect on wildlife and the potential harm they could cause to human beings, the use of most of these organochlorine pesticides, eg DDT, BHC (except the gamma isomer lindane), dieldrin, aldrin, endrin, and chlordane has been restricted or banned in most First World countries,³² (as well as South Africa) leading to the signing of the Stockholm Convention in 2004. In 1985, however, South African government officials used large quantities of stockpiled BHC against locusts, while the use of DDT as a method of malaria control was reintroduced in certain areas such as Kwazulu Natal, Limpopo and Mpumalanga.³³ Materials such as DDT and dieldrin are still used occasionally by farmers who (illegally) own or procure supplies; high levels of dieldrin were detected in melons in 1991,³⁴ for example, and a survey carried out in the Stellenbosch district during 2003 showed that 56 per cent of the farms were still in possession of obsolete pesticides,³⁵ despite the retrieval in 1999/2000 of some 1 050 tonnes of obsolete products (particularly from government storage) by Croplife SA and partners, an obsolete chemical retrieval and disposal programme funded by the Department of Agriculture. A positive case in 2007 of dieldrin poisoning of baboons, and a non-lethal poisoning of a veterinary worker affected by the vomitus of one baboon, indicate that despite being banned, illegal stocks of dieldrin still exist. An initiative to address the dangers posed by the accumulation of obsolete pesticide stockpiles not only in South Africa but throughout the African continent was proposed in 2000, leading to the creation of the Africa Stockpiles Programme (AFP). South

²⁹ London 'Human Rights, Environmental Justice, and the Health of Farm Workers in South Africa' 2003 *International Journal of Occupational and Environmental Health*.

³⁰ Mott & Snyder (1987) 18 *Pesticide Alert—A Guide to Pesticides in Fruit and Vegetables*.

³¹ <http://www.answers.com> 'Pesticide poisoning'

³² Anon 'Gebruik van verskeie insekmiddels ingetrek of ingekort' 1970 (674) *Agricultural News* 1.

³³ Wells & Leonard 'DDT Contamination in South Africa' (Groundwork May 2006).

³⁴ *Die Burger* 19 February 1991.

³⁵ Dalvie, Africa & London 'Disposal of Unwanted Pesticides in Stellenbosch, South Africa' 2006 (361) *Science of the Total Environment* 8.

Africa is one of seven countries in phase one of the programme, which commenced in 2007 and aims to clear all obsolete pesticide stocks from Africa over the next 10–15 years.³⁶

As a special case, DDT is still applied by Department of Health officials as an indoor residual spray (IRS) on the inside walls of huts in high risk malarial areas for the control of mosquitoes (transmitters of the malaria parasite). Studies conducted in KwaZulu-Natal revealed that levels of DDT in human serum³⁷ and breast milk³⁸ in areas where DDT is used for mosquito control are significantly higher than in non-control areas; breast milk from these areas also contained residues of pyrethroid insecticides³⁹, currently under scrutiny for potential carcinogenic effects in humans. The levels found in breast milk are considered a possible health risk to infants.⁴⁰ In certain areas fish also showed increased levels of DDT, particularly those higher up in the food web.^{41,42} This does not seem to pose a health hazard to the local population, but possible deleterious effects to species in higher trophic levels seem possible.⁴³

Many pesticides have recently been targeted as environmental endocrine disruptors. They act to mimic estrogen or act as antiestrogens or antiandrogens, are detrimental to reproduction and may promote abnormalities such as a decrease in sperm count and an increase in testicular cancer.⁴⁴ Pesticides investigated include the insecticides DDT, metoxychlor, and trichlorfon; the fungicide vinclozolin, and the herbicide atrazin.⁴⁵ In particular, the estrogenic nature of DDT and the anti-androgenic nature of its metabolite p,p'-DDE have been linked to a rise in reproductive abnormalities in both humans and wildlife over the last 20–30 years.⁴⁶

The fear that pesticides may be carcinogenic or at least oncogenic (inducing either benign or malignant tumours) has increased with the knowledge that many

³⁶ Information on the African Stockpiles Programme can be found at <http://www.africastockpiles.net/pressroom>.

³⁷ Bouwman *DDT Levels in Serum, Breast-milk and Infants in Various Populations in Malaria and Non-malaria Controlled Areas of KwaZulu* Medical Research Council (1991); Bouwman et al 'Malaria Control and Levels of DDT in Serum of Two Populations in KwaZulu' 1991 (33) *Journal of Toxicology and Environmental Health* 141.

³⁸ Bouwman above n 20; Bouwman, Reinecke, Cooppan & Becker 'Factors Affecting Levels of DDT and Metabolites in Human Breast-milk from KwaZulu.' 1990 (31) *Journal of Toxicology and Environmental Health* 93.

³⁹ Bouwman, Sereda & Meinhardt loc cit.

⁴⁰ Ibid.

⁴¹ Bouwman, Coetzee & Schutte 'Environmental and Health Implications of DDT-contaminated Fish from the Pongolo Flood Plain' 1990 (104) *South African Journal of Zoology* 215; Du Preez, Heath, Sandham, et al 'Risks Associated with the Consumption of Chemical Contaminated Freshwater Fish in South Africa' 2003 *Water South Africa* (Water Research Commission 2003.)

⁴² Mbongwe et al 'Dichlorodiphenyltrichloroethane in the Aquatic Ecosystem of the Okavango Delta, Botswana, South Africa' 2003 (22) *Environmental Toxicology and Chemistry* 7.

⁴³ Ibid.

⁴⁴ Anway, & Skinner 'Epigenetic Transgenerational Actions of Endocrine Disruptors' 2006 (147) *Endocrinology* 44.

⁴⁵ Ibid.

⁴⁶ Wells & Leonard loc cit

herbicides, insecticides, and fungicides have been shown to produce such effects in rodents in the United States. As a result of investigations, the Environmental Protection Agency (EPA) of the United States considered a substantial number of commonly used pesticides to be oncogenic (tumour producing) or potentially oncogenic in animal studies⁴⁷, which pesticides include many chemicals commonly used in South Africa today, eg the herbicides alachlor (Lasso) and paraquat (Grammoxone), the fungicides benomyl (Benlate) and mancozeb (Dithane M45), and the insecticides azinphos-methyl (Gusathion), cypermethrin (Cymbush, Ripcord), dicofol (Kelthane), gamma-BHC (Lindane), parathion (Folidol) and permethrin (Ambush).

Resistance to the use of pesticides harmful to human health and the environment has grown in Europe too, leading to the adoption by the European Parliament in January 2009 of a new regulation which establishes clear criteria for the approval of substances and stipulates that 22 commonly used pesticides be phased out over the next 10 years.⁴⁸

Certain toxicologists question the validity and value of rodent carcinogenicity studies for assessing the cancer risks to human beings posed by pesticides. They contend that tests in which near-toxic doses are applied to rodents do not provide enough information to predict the excess numbers of human cancers that might occur at low-dose exposures. Moreover, the quantity of synthetic pesticide residues in a normal diet is minute compared to that of natural pesticides, ie chemicals that plants use to defend themselves against insects and pathogens; furthermore, there is no fundamental difference between these groups of chemicals.⁴⁹ Other researchers found that carcinogenic effects were generally not limited to the top exposure level and were frequently observed at lower levels where there is no evidence of toxicity; in fact, increases in tumour rates are often present at lower doses.⁵⁰ A further consideration is that comparisons based strictly on the relative mass of synthetic and natural pesticides consumed are of little scientific value, because the carcinogenic potency of the various chemicals is not taken into account. It should also be borne in mind that many people are exposed to more sources of pesticides than just residues in food, particularly farmers, occupationally exposed workers, pesticide applicators, and weekend gardeners.^{51,52}

Whatever the case may be, it is very important that the presence of pesticide residues in food be monitored on a regular basis. In view of the great range of

⁴⁷ *Regulating Pesticides in Food* National Research Council, USA (1987) 20.

⁴⁸ <http://www.theparliament.com/latestnews/news-article/newsarticle/eu-parliament-backs-pesticide-regulations/>

⁴⁹ Ames & Gold 'Natural Plant Pesticides pose Greater Risks than Synthetic Ones' (1991) 69 *Chemical and Engineering News* 48; Ames & Gold 'Cancer Prevention Strategies Greatly Exaggerate Risks' 1991 (69) *Chemical and Engineering News* 28.

⁵⁰ Huff & Haseman 'Exposure to Certain Pesticides May Pose Real Carcinogenic Risk' (1991) 69 *Chemical and Engineering News* 33; Clapp, Jacobs & Loechler 'Environmental and Occupational Causes of Cancer: New Evidence:2005—2007' (2008) *Reviews on Environmental Health* 1.

⁵¹ Haseman & Huff 'Arguments that Discredit Animal Studies Lack Scientific Support' 1991 (69) *Chemical and Engineering News* 49.

⁵² London above n46.

pesticides, the variety of products in which they can occur, and the many marketing outlets for foodstuffs, it is difficult and expensive to do justice to the real requirements in this regard. Currently, routine analyses for certain pesticides are carried out on export fruits, but products on the local markets are tested for residues only on an ad hoc basis – ie certain products are tested for certain pesticides when samples are submitted by health inspectors.⁵³ Since it is almost impossible to control the way in which farmers use pesticides, ie whether they spray a registered pesticide at the registered dose and according to the prescribed safety period before marketing products, regular monitoring for residues on products that reach the market is the only way to ensure that consumers are not exposed to toxic chemicals. Some products do not pass through the big wholesale markets—where samples can easily be obtained—but are sold directly to super-markets and cafés or along roads in farm stalls, where special attempts should be made to determine whether farmers have used pesticides responsibly. Particular attention should be given to those products that may have required spraying during the extended periods over which the crop is harvested, such as strawberries and tomatoes, as well as to products sold directly to consumers by small and emerging farmers.

The health of pesticide applicators may give important indications of the effects of chronic exposure to moderate levels of pesticides. In South Africa farm workers who applied organophosphate insecticides experienced significantly more neurological symptoms such as dizziness, sleepiness, and headaches than non-applicators.⁵⁴ In the extensive Agricultural Health Study conducted in the beginning of the 21st century in the United States amongst pesticide applicators, it was also found that a prevalence of neurological symptoms was associated with lifetime exposure to pesticides, particularly organophosphate and organochlorine insecticides and fumigants.⁵⁵ Furthermore, indications were found of an association between chlorpyrifos, a commonly used organophosphate, and lung cancer.⁵⁶

19.3.2 Impact on the population level

Members of the same species that constitute a population have many inherent genetic differences, also regarding their response to a stress factor such as a pesticide application. In a large population of weeds, pathogens, or insects, there may be some individuals that can survive a generally effective dosage at the registered rate of application. Some individual insects may have superior qualities in metabolising the insecticide; in others the rate of penetration and transport of the chemical to the site of action may be reduced, while yet others may succeed in

⁵³ Barlin-Brinck 'Pesticides in Southern Africa – An Assessment of their Use and Environmental Impact' (1991).

⁵⁴ London et al 'Effects of Long-term Organophosphate Exposures on Neurological Symptoms, Vibration Sense and Tremor among South African Farm Workers' 1998 (24) *Scandinavian Journal of Work, Environment & Health* 18.

⁵⁵ Kamel et al 'Neurological Symptoms in Licensed Private Pesticide Applicators in the Agricultural Health Study' 2005 (113) *Environmental Health Perspectives* 877.

⁵⁶ Lee et al 'Cancer Incidence among Pesticide Applicators Exposed to Chlorpyrifos in the Agricultural Health Study' 2004 (96) *Journal of the National Cancer Institute* 1781.

detecting and avoiding (or are repelled by) insecticide-treated surfaces. Some are merely less sensitive. Those that survive spray applications because of their genetic characteristics will transmit these traits to the next generation. Through repeated selective action by subsequent sprays a resistant population can eventually develop. This phenomenon may be exacerbated by under application of a pesticide in a situation where the pesticide is over-diluted to save money. Previous reference was made to the lack of control over pesticide management by end users or over farmer activities.

By 1990 already more than 500 species of arthropod worldwide had developed resistance to one or more insecticides⁵⁷ and there are many more today.⁵⁸ In South Africa the problem was particularly severe in the cases of red spider mite and citrus red scale. Lack of effective acaricides and insecticides against these organisms has stimulated investigations into using biological control agents against these two pests, with remarkable success. In the case of codling moth, a severe pest in apples, resistance to insecticides has led to the development of synthetic sex pheromones that cause mating disruption. In view of the ability of organisms to adapt to new compounds – in some cases very quickly – and the high cost of developing new pesticides, it is realised that we may run out of replacement molecules. New strategies in managing pest numbers include mating disruption, the release of sterile males, using genetically modified plants and rotating available pesticides.

19.3.3 Impact on the community level

The different populations that constitute a community are affected differentially according to the general inherent susceptibility of the species to the dosage applied. Species of insects that have been exposed to a wider range of chemicals in their evolutionary history, such as those with a wide range of host plants, are better equipped to detoxify pesticides than the monophagous species.⁵⁹ One would also expect the carnivores (predators and parasitoids), which have not been exposed to the vast array of toxic chemicals in plants, to be generally less efficient in detoxifying chemicals than the herbivores. The implication of this is that the natural enemies (carnivores) of our pest organisms (mostly herbivores) are killed more easily than the pests themselves. Not only is the direct mortality caused by the pesticides important, but also the effect of sublethal doses which may interfere with normal behaviour patterns (eg host recognition, sexual communication) or the physiological processes of the natural enemies.⁶⁰ Where natural enemies have been important in depressing populations of pests or

⁵⁷ Georgioui 'Overview of Insecticide Resistance' in Green, Lebaron & Momberg (eds) *Managing Resistance to Agrochemicals: From Fundamental Research to Practical Strategies* ACS Symposium Series 421 (1990) 421.

⁵⁸ <http://www.pesticideresistance.com>

⁵⁹ Wilkinson 'Role of Mixed Function Oxidases in Insecticide Resistance to Pesticides' in Georgioui & Saito (eds) *Pest Resistance to Pesticides* (1983) 175, 191.

⁶⁰ Elzen 'Sublethal Effects of Pesticides on Beneficial Parasitoids' in Jepson (ed) *Pesticides and Non-target Invertebrates* (1989) 129; Stapel, Cortesero & Lewis 'Disruptive Sublethal Effects of Insecticides on Biological Control' (2000) 17 *Biological Control* 243.

potential pests, this impact of pesticides can result in the resurgence of pest numbers soon after a spray application or the emergence of new pests from the community.

In addition to their effect on natural enemies, pesticides may also kill other beneficial insects such as the pollinators on which most plants depend, including those of many commercial crops. They may also kill susceptible non-pest species, which in turn affect the species at the next trophic level (such as insectivorous birds) for which they have served as food.

Herbicides, like other pesticides, simplify communities through their selective action on populations – the so-called weeds and other susceptible species are removed, and with them the many insect species that feed on them. They therefore remove not only plants that provide shelter to animals but also a source of food for the herbivores and indirectly also for the carnivores.

19.3.4 Impact on the ecosystem level

Ecosystems are communities of interacting plants and animals together with their physico-chemical environment. Pesticides are not only found in the living component of ecosystems (as described above) but can also remain for some time in the air, water, and soil.

Aerosol cans containing insecticides to control flying insects are specifically designed so that small toxic droplets remain in the air for as long as possible. The insecticide therefore becomes part of the air that higher animals inhale. When crops are sprayed with pesticides, only minute quantities reach the target organisms—the rest contaminate the air, the soil and water systems. Depending on the type of pesticide, shorter or longer periods may elapse before it is degraded through physical, chemical, and biological processes. This rate of chemical degradation is expressed as a half life figure, or the time taken for half of the volume of active ingredient to decompose to environmentally inactive components.

Most pesticides are insoluble in water, but since water is the most commonly used diluent and carrier from the sprayer to the target, they are formulated in such a way that they form suspensions or emulsions when mixed with water. This facilitates their dispersal in soil, groundwater and streams. They are also carried with the organic and clay particles to which they adsorb, while some of them can move as organic vapour through the unsaturated zone of the soil. Detailed descriptions of the processes involved are given by Yaron.⁶¹ The result is that these toxic substances are eventually found far beyond the area where they were applied. Studies have shown that pesticides can move through the soil much faster and to greater depths than previously expected.⁶² Information on the behaviour of pesticides in soil is summarised by Saltzman and Yaron.⁶³

⁶¹ Yaron 'General Principles of Pesticide Movement to Groundwater' 1989 (26) *Agriculture, Ecosystems and Environment* 275; Pimentel *Encyclopedia of Pest Management* (CRC Press 2007).

⁶² Hallberg 'Pesticide Pollution of Groundwater in the Humid United States Agriculture' 1989 (26) *Agriculture, Ecosystems and Environment* 299, 348.

⁶³ Saltzman & Yaron *Pesticides in Soil* (1986).

The presence of pesticides in groundwater is cause for international concern. Studies conducted more than 20 years ago showed that pesticides leach through the soil much more commonly than historic preconceptions predicted.⁶⁴ This applies also to the post-DDT chemicals, which were thought to be safer because they would degrade relatively rapidly in the topsoil. It was suggested they would be adsorbed to soil particles for long enough to degrade. Inadequate research resulted in bioaccumulation and toxicity being higher than predicted.⁶⁵ Pesticides commonly found in groundwater were the herbicides alachlor and atrazine, the insecticides aldicarb and carbofuran (also used against nematodes) and the nematicides DBCP and EDB. Typical concentrations ranged from 0,1 to 20/ $\mu\text{g.l}^{-1}$. Some of these pesticides have also been found in groundwater in Europe, where the Council of the European Communities established a maximum admissible concentration of 0,1/ $\mu\text{g.l}^{-1}$ for any pesticide in drinking water.⁶⁶ This value represents a detection limit in chemical analysis: the idea is that there should be no pesticide residues in drinking water. In the Netherlands, the use of certain pesticides is not allowed in proclaimed groundwater protection areas.⁶⁷

In the Lourens River near Somerset West, relatively high concentrations of the insecticides azinphos-methyl and endosulfan (up to 1,5 $\mu\text{g.l}^{-1}$ and 2,9 $\mu\text{g.l}^{-1}$ respectively) were found downstream of an orchard after rain. These levels are considered to be toxic to fish and macro-invertebrates and greatly exceed the national water quality standard of <0.01 $\mu\text{g.l}^{-1}$ set by the South African Department of Water Affairs and the US EPA.⁶⁸ In the same river chlorpyrifos was found in concentrations of up to 344 $\mu\text{g.l}^{-1}$ in suspended sediments after the first rains that followed the termination of spray applications. This level already poses a potential hazard to sediment-dwelling organisms,⁶⁹ but after heavy rains during the spraying period, levels as high as 924 $\mu\text{g.l}^{-1}$ were measured.⁷⁰ This chemical persists in the soil for up to six months after a spray application at concentrations that adversely affect earthworms, a non-target organism.^{71,72} The fungicides dithiocarbamate and copper oxychloride were also shown to have

⁶⁴ Hallberg op cit 357; Foster & Chilton 'Mechanisms of Groundwater Pollution by Pesticides' (2007) 5 Issue 2 *Water and Environmental Journal* 186-193.

⁶⁵ Hallberg op cit 300.

⁶⁶ Leistra & Boesten 'Pesticide Contamination of Groundwater in Western Europe' 1989 (26) *Agriculture, Ecosystems and Environment* 369, 370.

⁶⁷ Leistra & Boesten op cit 369.

⁶⁸ Schulz 'Comparison of Spray Drift- and Runoff-related Input of Azinphos-methyl and Endosulfan from Fruit Orchards into the Lourens River, South Africa' 2001 (45) *Chemosphere* 543.

⁶⁹ Schulz et al 'Current-use Insecticides, Phosphates and Suspended Solids in the Lourens River, Western Cape, During the First Rainfall Event of the Wet Season'. 2001 (27) *Water SA* 65.

⁷⁰ Schulz 'Rainfall-induced Sediment and Pesticide Input from Orchards into the Lourens River, Western Cape, South Africa: Importance of a Single Event' 2001 (35) *Water Research* 1869.

⁷¹ Reinecke & Reinecke 'Biomarker Response and Biomass Change of Earthworms Exposed to Chlorpyrifos in Microcosms' 2007 (66) *Ecotoxicology and Environmental Safety* 92.

⁷² Reinecke & Reinecke 'The Impact of Organophosphate Pesticides in Orchard on Earthworms in the Western Cape, South Africa' 2007 (66) *Ecotoxicology and Environmental Safety* 244.

negative impacts on earthworms and their activities, affecting dispersal, establishment, and population development.^{73,74}

19.4 REDUCING IMPACTS THROUGH INTEGRATED PEST MANAGEMENT

Concern over the widespread and increasing occurrence of resistance, as well as health and environmental hazards associated with the use of pesticides has prompted scientists to develop pest control strategies that are less reliant on pesticides. The aim is to move away from routine or calendar spraying, where applications are made at regular intervals or at fixed periods in the development of the crop, regardless of infestation levels. Instead, pesticides are applied only when the numbers of pest insects exceed predetermined thresholds that may result in economic damage. At the same time, conditions are created that are unfavourable for the pest in an effort to keep its numbers below the threshold. Thus biodiversity on farms is enhanced. Resistant crop plants are bred through selection or gene transfer (genetic modifications) and used as far and for as long as possible. The effect of the pests' natural enemies is enhanced by introducing new species, using selective sprays that are 'soft' on them and augmenting their numbers by rearing and releasing; sanitary measures are applied to destroy infested plant residues or fruit; sticky materials are used to trap or exclude insects, and sex pheromones are released, which disrupts mating.

This approach is known as Integrated Pest Management (IPM). It requires, of course, that pest organism numbers are monitored regularly and that decisions are made on the appropriate remedial action. It is also dependent on the availability of reliable information on the damage potential of pests, their rate of development and the potential of natural enemies to suppress them. Successful IPM therefore depends on sophisticated biological research and competent management inputs.

In South Africa, the greatest progress with IPM has been made in the cases of citrus and cotton, and the number of pesticide sprays applied has been reduced considerably.⁷⁵ For many other major pest species such as codling moth and spider mites on deciduous fruit, the maize stalk borer and the Russian wheat aphid, maximum thresholds have been determined for pest numbers before spraying is considered necessary. In many cases, however, these levels are regularly reached because no effective biological or other non-chemical control measures are available. Also, many crops have a range of pests and diseases attacking them and when sprays have to be applied for one pest (eg citrus thrips), this may upset the biological control of other pests (eg red scale) on the same trees.

This illustrates the dynamic nature of pest management and the need for

⁷³ Vermeulen, Reinecke & Reinecke 'Evaluation of the Fungicide Manganese-zinc ethylene Bis (dithiocarbamate) (Mancozeb) for Sublethal and Acute Toxicity to *Eisenia fetida* (Oligochaeta) 2001 (48) *Ecotoxicology and Environmental Safety* 183.

⁷⁴ Eijssackers et al 'The Implications of Copper Fungicide Usage in Vineyards for Earthworm Activity and Resulting Sustainable Soil Quality' 2005 (62) *Ecotoxicology and Environmental Safety* 99.

⁷⁵ Annecke & Moran *Insects and Mites of Cultivated Plants in South Africa* (1982) 35, 254.

continued research to produce the information on which management decisions can be based. Many farmers are reluctant to follow IPM because they lack the management expertise and are not prepared to risk producing lower-quality fruit as a result of incorrect monitoring of pest numbers or wrong decisions. Other restraints are the possibility of bad weather or insufficient equipment if spraying is delayed until threshold levels are reached.⁷⁶ The result is that they apply pesticides as a precaution, whether required or not, and ignorance often leads to over-applications 'just in case'.

Towards the end of the 1980s a movement was started in Europe to sell deciduous fruit under a special label if produced with environmentally compatible methods.⁷⁷ An example is the Euro-Retail Produce Working Group programme to develop Good Agricultural Practices, known as EurepGAP. To meet these requirements strict guidelines were set up by local government agencies or farmers' co-operatives on various aspects of fruit production, including the use of chemicals such as fertilisers or pesticides, and adherence to these guidelines is supervised by auditors. This has developed as a trend internationally, where environmentally responsible production builds a market niche and finds buyers that are prepared to give preference to such produce, thus rewarding farmers for their efforts in protecting the environment and consumer health.⁷⁸ South African fruit growers have produced their own guidelines for what is now called Integrated Fruit Production. Another initiative in this regard is the South African Pesticide Initiative Program (SA PIP), co-funded by the European Commission and other stakeholders, which aims to assist the South African producers of horticultural exports to comply with EU requirements for food safety and consumer protection, in particular with regard to pesticide regulations, and thereby ensure continued access to the EU market for South African fruit and vegetables.⁷⁹

Many consumers are not happy to buy fruit with even the minimum residue levels (MRLs) set by the Department of Health and also demand food items completely free of pesticide residues. This can be achieved only by applying chemicals that break down rapidly long before harvest, and strictly according to manufacturers' directions, or by organic production – that is production without the use of synthetic pesticides and fertilisers. The market for organic products is growing rapidly and is backed by the supermarkets and ever-increasing consumer demand for healthy food. With good management, organic production can be competitive and may be more sustainable in the long term.⁸⁰

The majority of pesticide companies in South Africa are aware of their chemical stewardship obligations and most companies active in the field belong

⁷⁶ Giliomee 'Integrated Pest Management in Apple Orchards: Where Do we Stand?' 1989 (85) *South African Journal of Science* 361.

⁷⁷ Oberhofer 'Integrated Apple Production in Western Europe' *Cape Pomological Association Symposium Proceedings* (1990) 17.

⁷⁸ Giliomee 'A Green Perspective on Fruit Production' *Cape Pomological Association Symposium Proceedings* (1990) 49.

⁷⁹ <http://www.ppecb.sapip/>

⁸⁰ *Organic Agriculture: Sustainability, Markets and Policies* Organisation for Economic Co-operation and Development 2003.

to the Agricultural and Veterinary Chemicals Association of South Africa (AVCASA). However, some of these pay lip-service to environmental issues and merely conform to minimum legal requirements. Others have dedicated stewardship programmes in place. A certificate course is a general pre-requisite for field advisors who are required to carry an identification card, but whether this aspect is enforced or not remains open to debate. The Aerial Application Association of South Africa conducts annual 'fly-ins' for the calibration and certification of crop-spraying aircraft. In general, the levels of safety training for farm workers handling and applying pesticides have been found to be disturbingly low,⁸¹ with often inadequate forms of protection available.⁸² There is also great concern about the lack of awareness about pesticide risks amongst the growing class of emerging farmers who are being encouraged to engage in high-input agricultural production in order to obtain commercial status. Since existing practices in commercial agriculture rely heavily on pesticides, emerging farmers aspiring to become commercial are pressured to adopt and/or increase the use of pesticides, with the result that there is evidence that these farmers engage in high-risk practices such as the dangerous handling and improper disposal of pesticides.⁸³

19.5 CONTROL LEGISLATION

Many different Acts have a bearing on the use of pesticides in South Africa.⁸⁴ The most important is the Fertilizers, Farm Feeds, Agricultural Remedies and Stock Remedies Act 36 of 1947, which still regulates the importation, sale, acquisition, disposal, and use of pesticides, amongst other things. This Act is in the process of revision and as a first step in that direction a draft pesticide management policy for South Africa was published for public comment during 2006.⁸⁵ Some of the objectives of the policy are to minimise the hazards and risks to health and the environment, to improve the legislative framework for the use and distribution of pesticides, and to encourage the use of alternative products and techniques that reduce dependence on chemical plant protection products.

One of the main stipulations in the present Act is that pesticides have to be registered and for that purpose the Minister of Agriculture has to appoint an officer in his or her department as the Registrar. The Registrar must be satisfied that the remedy for which registration application is made is suitable and sufficiently effective for the purpose for which it is intended, that registration will not be contrary to the public interest and that the establishment where the remedy is manufactured is suitable. The Registrar may also cancel a registration if he or she is satisfied that it is contrary to the public interest that a product remains registered. No product may be sold unless it is registered and packed as prescribed. The Act is aimed principally at the protection of agriculture and seeks

⁸¹ London 'Agrichemical Safety Practices in the Western Cape' 1994 (84) *South African Medical Journal* 277.

⁸² London above n46.

⁸³ Rother, Hall & London op cit 420.

⁸⁴ Rother, Hall & London op cit 402.

⁸⁵ *Government Gazette* No. 28711 13 April 2006 80.

to ensure that only approved pesticides reach the consumer.⁸⁶ More than 3 000 pesticides have been approved for use in South Africa.⁸⁷ In terms of the Act, pest control operators also have to be registered and they may not use any product in a manner other than specified on the label. A South African Pest Control Association exists, and as with most such associations, the leadership echelon are compliant with legislation. Of concern are the suburban 'fly-by-night' pest controllers who operate without adequate training, competence or regulation.

One of the limitations of Act 36 of 1947 is that it does not require a regular review of registered pesticides.⁸⁸ The new policy, which should be implemented by way of revision or replacement of this Act, stipulates that special attention be given to pesticides that pose an unmanageable risk. Such pesticides should be considered for phase-out, severe restriction and bans. This will include those with Endocrine Disrupting Properties (EDPs), Persistent Organic Pollutants (POPs), formulations classified by the World Health Organization as Extremely Hazardous and Highly Hazardous, (WHO Group 1a and 1b) as well as pesticides which are repeatedly implicated in poisoning incidents.⁸⁹ In compliance with the Stockholm Convention on Persistent Organic Pollutants, adopted in 2001, South Africa has banned all the POPs listed under the Convention, with the exception of DDT, which is permitted only for government use in malaria vector control, and subject to conditions imposed by the Stockholm Convention.⁹⁰ South Africa is also a party to the Rotterdam Convention, which came into force in 2004 for the purpose of promoting shared responsibility and cooperative efforts among parties in the international trade of certain hazardous chemicals and to contribute to the environmentally sound use of those hazardous chemicals. This is achieved by facilitating information exchange about their characteristics, providing for a national decision-making process on their import and export and by disseminating these decisions to parties. One of the key provisions of the Rotterdam Convention is the Prior Informed Consent (PIC) procedure, whereby parties undertake, amongst others, to inform a pesticide-importing country when they plan to export a chemical that is banned or severely restricted for use within its territory.⁹¹ The procedure provides all parties with an opportunity to make informed decisions as to whether they will consent to future imports of the chemicals listed in Annex III of the Convention. All parties are required to ensure that their exports do not take place contrary to an importing party's import decision.

Preventing dangerous levels of pesticide residues in food is an objective of the Foodstuffs, Cosmetics and Disinfectants Act 54 of 1972. Regulations make provision for the establishment of maximum residue limits (MRLs) that may be present in foodstuffs when they reach the consumer. South Africa is a member of the Codex Alimentarius Commission (WHO/FAO), created in 1963 under the Joint FAO/WHO Food Standards Programme to develop food standards,

⁸⁶ Giliomee & Glavovic op cit 536.

⁸⁷ *Government Gazette* op cit 83.

⁸⁸ *Government Gazette* op cit 84.

⁸⁹ *Government Gazette* op cit 86.

⁹⁰ *Government Gazette* op cit 90.

⁹¹ <http://www.pic.int/>

guidelines and related texts such as codes of practice. The Codex Alimentarius (CA) attempts to set internationally acceptable MRLs on edible commodities which are traded internationally and CA standards are recognized by the World Trade Organization as critical reference points for international trade. As a recent example, in 2006 the CA adopted MRLs for sulfuryl fluoride as an alternative to the soil fumigant methyl bromide, which is being phased out due to its ozone-depleting characteristics. In South Africa the goal is to phase out about 1000 tonnes of methyl bromide by 2015 and by 2004 had taken many regulatory steps to reduce consumption.⁹²

The Hazardous Substances Act 15 of 1973 makes provision for the Minister of Health to declare very toxic pesticides (to human beings) as Group 1 hazardous substances. Regulations promulgated in terms of this Act regulate the licensing, conditions of sale, records to be kept by licensees, and the labelling and disposal of empty containers of such pesticides. The Act is clearly aimed at the protection of public health. Poisoning from any registered pesticide is a notifiable medical condition in terms of the Health Act 63 of 1977.

The Occupational Health and Safety Act 85 of 1993 makes provision for various mandatory safety measures to protect the health of workers handling pesticide, such as risk assessment, safety training, safe practices, and medical and environmental monitoring of workplaces.

19.6 CONCLUSIONS

It is clear that when pesticides are applied against pest organisms, they are in fact impinging on highly complex ecosystems, affecting individuals, populations, communities, and ecosystems in many different ways. In so far as they reduce the numbers of harmful organisms in crop and animal production, as well as disease vectors, they are of great benefit to humankind. However, their use and persistence in the environment should continuously be monitored so that the effects of the ecological disturbance and possible hazards they cause can become known and be prevented. Of particular concern is the potential danger of pesticide residues in food such as fruit and vegetables where the chemical is applied directly to the edible part of the plant. Since there is and can be no effective direct control over the way farmers use pesticides, this should be done indirectly by extensive monitoring of their products, not only those reaching the big export markets but also those that are sold in local supermarkets and smaller outlets such as farm stalls.

There are a number of alarming shortcomings in the South African pesticide regulatory and legislative systems. Amongst these is the lack of control over pesticide sales. The end user responsibility is seriously lacking, and control and management of pesticides frequently so lackadaisical that ignorant, negligent or reckless farmers are not exposed and penalised. The lack of knowledge about the extent of pesticide levels in fruit and vegetables sold on the formal and informal markets is also alarming. Certain pesticides are difficult to identify by routine

⁹² Gonzalez 'Information on the Request of South Africa for Assistance from GEF to Phase out Methyl Bromide' April 2004 www.gefweb.org

laboratory methods, and maximum pesticide residue levels officially allowed in food are poorly publicised and monitored.

Exacerbating these problems is the tendency for promotional marketing by the chemical industry, where purchase incentives are offered to farmers, and sales incentives to outlets and sales representatives. The result is the accumulation of pesticide supplies that later become expired, obsolete or unwanted, and which are then difficult and extremely expensive to retrieve and dispose of.

More than 20 years ago a report by the EPA ranked pesticides as one of the most serious health and environmental problems in the United States.⁹³ The concerns expressed in this report are still relevant today⁹⁴ and the South African authorities are urged to take note thereof.⁹⁵ In particular, the work of determining base levels of pesticides in our general environment and in food in particular should be extended, with speedy action taken when pesticides are misused or the normal use causes unacceptable risks and environmental damage. The Department of Environmental Affairs and Tourism (DEAT) is currently in the process of compiling a National Chemicals Profile for South Africa, with the aim to comprehensively and systematically assess the country's existing chemicals management infrastructure and capabilities. According to the DEAT, the National Profile will highlight the national priorities and identify existing gaps and weaknesses in the chemicals management infrastructure. From this information, the Department will then draft the national action plan to develop an integrated approach for chemicals management.⁹⁶

Over the years, the pesticide industry has been making great progress in developing chemicals that are very effective, yet are less toxic to humans, less persistent in the environment and more selective in targeting organisms. Government policies, financial incentives and pressure from the anti-pesticide lobby will hopefully ensure that we continuously move forward in this regard, as pesticides are likely to be an essential component in agricultural production and vector control for many years to come. Initiatives such as the Strategic Approach to International Chemicals Management (SAICM),⁹⁷ adopted in 2006 by the Intergovernmental Forum on Chemical Safety should be actively supported. As a policy framework, the SAICM aims to ensure that chemicals are produced and used in ways that minimise significant adverse impacts on the environment and human health and should result in policies, legislation and procedures that foster the sound management of chemicals internationally. As such, this will allow for responsible use by farmers and pest control operators of chemicals that are of benefit in protecting us from organisms that threaten our food supply, health and the environment generally.

⁹³ Mott & Snyder op cit 17, 18.

⁹⁴ Kortenkamp, Faust, Scholze & Backhaus 'Low-Level Exposure to Multiple Chemicals: Reason for Human Health Concerns?' (2007) 115 *Environmental Health Perspectives* 106–114.

⁹⁵ London & Rother 'People, Pesticides, and the Environment: Who Bears the Brunt of Backward Policy in South Africa?' (2000)10 *New Solutions: A Journal of Environmental and Occupational Health Policy* 339.

⁹⁶ <http://www.deat.gov.za>.

⁹⁷ <http://www.saicm.org>